Technology Stack

Date	June 2026
Team ID	xxxxxx
Project Name	RISING WATERS
Maximum Marks	2 Marks

Define Your Technology Stack:

The Rising Waters project uses a modern and reliable technology stack to develop a secure, scalable, and user-friendly flood monitoring and alert system. The frontend is built using HTML5, CSS3, JavaScript, and Bootstrap to create a responsive interface. The backend uses Python (Flask) to process requests and manage business logic. MySQL is used to securely store user details, flood data, and alerts. The application is hosted on Render/Railway for reliable deployment and scalability. Git and GitHub are used for version control and team collaboration. OpenWeather API, Google Maps API, and Twilio API are integrated to provide real-time weather information, interactive flood maps, and emergency SMS notifications.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	**Frontend / Client-Side**	HTML,CSS3,JavaScript, Bootstrap	Creates a responsive and user-friendly interface that works on desktop and mobile devices.

2	**Backend / Server**	Python (Flask)	Handles user requests, flood data processing, and API integration efficiently.
3	**Database / Data Storage**	MySQL	Stores user information, flood records, alerts, and historical data securely.
4	**Cloud / Hosting / Deployment**	Render / Railway	Provides reliable hosting with easy deployment and scalability.
5	**Version Control**	Git & GitHub	Tracks code changes, supports collaboration, and maintains project versions.
6	**Third-Party APIs / Other Tools**	Open Weather API, Google Maps API, Twilio	OpenWeather API, Google Maps API, Twilio OpenWeather API provides weather and rainfall data, Google Maps API displays flood-prone locations, and Twilio sends SMS emergency alerts to users.